Goodness of Estimators

Learning Outcomes

- Rao-Blackwell Theorem
- MVUE
- MLE Properties

Rao-Blackwell Theorem

Let the joint distribution of $X_1, \ldots, X_n$ depend on parameter θ and let T be a sufficient statistic for θ. If we consider estimating $h(\theta)$, a function of θ. If U is an unbiased estimator of $h(\theta)$, then the estimator $U^* = E(U|T)$ is also an unbiased estimator of $h(\theta)$ and has a variance no larger than U.

$$\text{Normal}(\mu, \sigma^2)$$

$$T = \begin{pmatrix} \sum X_i \\ \sum X_i^2 \end{pmatrix}$$

$$\hat{\mu}_{MLE} = \bar{X} = \frac{1}{n} \sum X_i$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{n} \sum (X_i - \bar{X})^2$$

$$\frac{1}{n}\left(\sum X_i^2 - n\bar{X}^2\right)$$

Minimum Variance Unbiased Estimator

Let $\mathcal{U}$ be a set of all unbiased estimators T of $\theta \in \Theta$ and $E(T^2) < \infty$. An estimator T_0 is said to be the minimum variance unbiased estimator for θ if

$$E\{(T_0 - \theta)^2\} \leq E\{(T - \theta)^2\}$$

for all $\theta \in \Theta$ and every $T \in \mathcal{U}$.

Invariance Property

If $\widehat{\theta}$ is an ML estimator of θ, then for any one-to-one function g, the ML estimator for $g(\theta)$ is $g(\widehat{\theta})$.

Large Sample Theory

Let $X_1, \ldots, X_n$ be a random sample from a distribution with parameter θ. Let $\widehat{\theta}$ be the MLE estimator for a parameter θ. As $n \to \infty$, then $\widehat{\theta}$ has a normal distribution with mean θ and variance $1/nI(\theta)$, where

$$I(\theta) = E\left[-\frac{\partial^2}{\partial\theta^2}\log\{f(X;\theta)\}\right]$$

Normal Distribution

Let $X_1, \dots, X_n$ be a random sample with the following distribution:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

Find the Information of μ.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\ln f(x) = -\frac{1}{2}\ln(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}$$

$$\frac{\partial f}{\partial \mu} = -\frac{1}{2\sigma^2} \cdot 2(x-\mu)(-1)$$

$$= \frac{(x-\mu)}{\sigma^2}$$

$$\frac{\partial^2 f}{\partial \mu^2} = -\frac{1}{\sigma^2}$$

$$E\left(\frac{1}{\sigma^2}\right) = \frac{1}{\sigma^2}$$

$$I(\mu) = \frac{1}{\sigma^2}$$

$$\bar{X} \sim N\left(\mu, \frac{1}{n\frac{1}{\sigma^2}} = \frac{\sigma^2}{n}\right)$$

Bernoulli Distribution

Let $X_1, \ldots, X_n$ be a random sample with the following distribution:

$E(X) = p$

$$f(x) = p^x (1-p)^{1-x}$$

Find the Information of p.

$$E\left(\frac{x}{p^2} + \frac{(1-x)}{(1-p)^2}\right)$$

$$\ln\left(p^x (1-p)^{1-x}\right) = \ln p^x + \ln (1-p)^{1-x}$$

$$E\left(\frac{x}{p^2}\right) + E\left(\frac{1-x}{(1-p)^2}\right)$$

$$\frac{1}{p^2} E(x) + \frac{1}{(1-p)^2} E(1-x)$$

$$\frac{p}{p^2} + \frac{1}{(1-p)^2}\left[E(1) - E(x)\right]$$

$$\frac{\cancel{p}}{p^{\cancel{2}}} + \frac{\cancel{1-p}}{(1-p)^{\cancel{2}}} = \frac{1}{p} + \frac{1}{(1-p)}$$

$$\frac{1}{p(1-p)} \qquad \hat{p}$$

$$n \to \infty$$

$$\hat{p} \sim N\left(p, \frac{1}{n \frac{1}{p(1-p)}} = \frac{p(1-p)}{n}\right)$$

Poisson Distribution

Let $X_1, \ldots, X_n$ be a random sample with the following distribution:

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

Find the Information of $\cancel{x}$ λ

$$\ln f(x) = \ln\left(\frac{e^{-\lambda}\lambda^{x}}{x!}\right)$$

$$= -\lambda + x\ln\lambda - \ln(x!)$$

$$\frac{df}{d\lambda} = -1 + \frac{x}{\lambda}$$

$$\frac{d^2f}{d\lambda^2} = -\frac{x}{\lambda^2}$$

$$E\left(\frac{x}{\lambda^2}\right)$$

$$\frac{1}{\lambda^2} E(X) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$$

$$I(\lambda) = \frac{1}{\lambda}$$

$$\hat{\lambda}_{MLE} \overset{n \to \infty}{\sim} N\left(\lambda, \frac{1}{n \frac{1}{\lambda}} = \frac{\lambda}{n}\right)$$

$$\hat{\lambda}_{MLE} = \bar{X}$$

Exponential Distribution

Let $X_1, \ldots, X_n$ be a random sample with the following distribution:

$E(x) = \theta$

$$f(x) = \frac{1}{\theta} e^{-x/\theta} \text{ for } x > 0$$

Find the Information of θ.

$$\ln f(x) = -\ln\theta - \frac{x}{\theta}$$

$$\frac{df}{d\theta} = -\frac{1}{\theta} + \frac{x}{\theta^2}$$

$$= +\frac{1}{\theta^2} - \frac{2x}{\theta^3}$$

$$= E\left(-\frac{1}{\theta^2} + \frac{2x}{\theta^3}\right)$$

$$-\frac{1}{\theta^2} + \frac{2}{\theta^3} E(x)$$

$$-\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = -\frac{1}{\theta^2} + \frac{2}{\theta^2}$$

$$I(\theta) = \frac{1}{\theta^2}$$

$$\hat{\theta}_{MLE} \overset{n \to \infty}{\sim} N\left(\theta, \frac{1}{n\frac{1}{\theta^2}} = \frac{\theta^2}{n}\right)$$